

Rishabh Budhouliya

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TECHNICAL SKILLS

Languages: Python, Rust, Go, Java, SQL

Systems: POSIX syscalls, WebAssembly / wasmtime, IPC, shared memory and mmap, sandboxing, B+ trees, storage engines, PyO3 / FFI

Tools: gdb, py-spy, memray, Git, Docker, Kubernetes, CI/CD, Google Cloud Platform, Amazon Web Services

EXPERIENCE

Systems Research Engineer

Jan 2026 – Present

Lind-Wasm – NYU Secure Systems Lab

New York, NY

- Got [PostgreSQL running inside a WebAssembly sandbox](#), adding static and dynamic build support and taking it through the pg_regress suite
- Implemented and extended the POSIX syscall surface in RawPOSIX, the Rust microvisor backing the sandbox: openat, mknod, fcntl64, recvfrom, waitpid, mmap and munmap
- Led migration of the filesystem namespace model from a LIND_ROOT path-prefix scheme to chroot, updating the microvisor and full test suite

Fellow

May 2025 – Aug 2025, Jul 2026 – Present

Recurse Center

New York, NY

- Built a [SQLite-inspired storage kernel](#) from scratch in Go, achieving 12x performance improvement through zero-copy reads, MMAP paging, and B+ tree optimization; outperformed SQLite by 33% on 1M row queries

Software Engineer II

Jan 2021 – May 2024

Narvar

Bangalore, India

- Built a unified shipping label API that cut client integration time from 10-15 days to **7 days** by abstracting carrier API complexities, implementing OAuth2.0, and centralizing multiple services behind a single secure endpoint
- Led the OAuth2.0 migration for the label generation system when UPS, the largest carrier partner, upgraded its infrastructure, shipping with zero downtime

Software Engineer

August 2020 – Jan 2021

Guavus

Gurugram, India

- Wrote Spark structured streaming and batch jobs to aggregate hourly, daily, and monthly network data

Contributor at Jenkins

August 2019 – May 2020

Google Summer of Code

Remote

- Optimized [Git Plugin](#) performance by benchmarking CLI vs Java git implementations, cutting job execution time by up to **50%** across **225,000+** Jenkins servers worldwide

EDUCATION

New York University

New York, NY

Master's in Computer Engineering

Aug. 2024 – May 2026

Thapar University

Patiala, India

Bachelor of Computer Science, Minor in Business Administration

July 2015 – July 2020

PROJECTS

atomic-pipe - Atomic Multi-Writer Pipe Messaging | *Rust*

2026

- Designed a framing protocol over POSIX pipes, sized against PIPE_BUF so concurrent writes from multiple processes are never interleaved by the kernel, with per-stream reader-side demultiplexing

ring-buffer - MPMC Ring Buffer over Shared Memory | *Rust, POSIX*

2026

- Built a multi-producer multi-consumer ring buffer over POSIX shared memory using raw libc (shm_open, ftruncate, mmap) with a byte-packed cursor header

BPE Tokenizer - Profiling-Driven Optimization | *Python, Rust, PyO3*

2026

- Built a byte-pair-encoding tokenizer from scratch, replacing the per-merge full-corpus rescan with a pair-to-words inverted index so each merge only touches affected words
- Ported the training loop to Rust via PyO3, crossing the FFI boundary once per train() call: **75x faster** than the original naive implementation, producing byte-identical vocab and merge order